

Optimal Execution: KKT-Based Allocation Algorithm

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I consider the problem of splitting a fixed parent order of size S into minute-by-minute child trades x_t over $T = 390$ intervals. Let $g_t(x)$ be the temporary impact per share at time t , so the total cost is

$$C(\{x_t\}) = \sum_{t=1}^T x_t g_t(x_t), \quad \sum_{t=1}^T x_t = S, \quad x_t \geq 0.$$

Piecewise Impact Function Following a piecewise model,

$$g_t(x) = \begin{cases} c, & 0 \leq x \leq D_t, \\ a_t x^p, & x > D_t, \end{cases} \quad a_t = \frac{c}{D_t^p}, \quad 0 < p < 1,$$

where c , D_t , and p are estimated from data.

KKT Conditions and Inversion Introducing a Lagrange multiplier λ for the equality constraint, stationarity implies

$$\frac{d}{dx}[x g_t(x)] = \lambda.$$

Since

$$\frac{d}{dx}[x g_t(x)] = \begin{cases} c, & x \leq D_t, \\ a_t(p+1)x^p, & x > D_t, \end{cases}$$

we invert to obtain

$$x_t(\lambda) = \begin{cases} D_t, & \lambda \leq c, \\ \max\{D_t, (\frac{\lambda}{a_t(p+1)})^{1/p}\}, & \lambda > c. \end{cases}$$

Bisection Algorithm

1. Compute D_t array, scalar c, p , and $a_t = c/D_t^p$ from data.

2. If $S \leq \sum_t D_t$, allocate

$$x_t = \frac{D_t}{\sum_u D_u} S$$

and stop.

3. Otherwise set $\lambda_{\min} = c$, $\lambda_{\max} = (p+1) \max_t a_t S^p$.

4. For $k = 1 \dots N$: let $\lambda = \frac{1}{2}(\lambda_{\min} + \lambda_{\max})$. If $\sum_t x_t(\lambda) > S$ then $\lambda_{\max} = \lambda$, else $\lambda_{\min} = \lambda$.

5. Return $x_t = x_t(\frac{1}{2}(\lambda_{\min} + \lambda_{\max}))$ which satisfies $\sum_t x_t \approx S$.

Analysis of Allocation Results Applying the algorithm with $S = 10,000$ produced the vector:

$$x_t \approx [12.97, 13.00, 13.03, \dots, 112.55],$$

which closely follows the shape of D_t . Since

$$S > \sum_{t=1}^T D_t,$$

the solution enters the tail regime and effectively scales the flat-region capacity by a factor

$$k = \frac{S}{\sum_{u=1}^T D_u} > 1$$

ensuring $\sum_t x_t = S$. The schedule is lowest at market open, rises in mid-day as displayed depth increases, dips slightly in the afternoon, and spikes at close where D_t peaks—confirming that trading more when liquidity is deepest minimizes total price impact.

Discussion and Rationale This KKT–bisection method was chosen because:

- *Data efficiency*: works with limited data and known $g_t(x)$, no ML training required.
- *Convex separability*: reduces to a 1-D root solve on λ .
- *Deterministic setup*: fixed S and pre-calibrated impact.
- *Transparency*: closed-form inversion and bisection yield clear optimality.

Advanced methods like stochastic control or RL are better suited when price drift, uncertainty in $g_t(x)$, or complex risk constraints come into play. For a static, separable cost and fixed order size, the KKT–bisection approach is both rigorous and computationally lightweight.